

A superlinear barrier for bounded-overlap row corrections

Asymptotic construction-method note

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This is a consequence of the one-column correction theorem in `../sparse_row_correction_barrier.tex`. It does not apply to arbitrary dice without the stated correction representation.

Theorem 1. *Let $t_1, \dots, t_K \in [0, 1]$ be distinct, and suppose $p_j(q) = t_q + \delta_j(q)$, $1 \leq j \leq m$, has uniform-interval mixed moments*

$$K^{-1} \sum_q \prod_{j \in S} p_j(q) = \frac{1}{|S| + 1}.$$

Suppose each row has nonzero corrections in at most d columns. There is an absolute $c > 0$ such that

$$K \geq c \min \left\{ m^2, \frac{m^{4/3}}{d^{4/3}} \right\}.$$

In particular any fixed bound on the number of corrected columns per row requires $K = \Omega(m^{4/3})$. For $d = 1$ the stronger bound $K = \Omega(m^2)$ holds.

Proof. Join two columns by an edge if some row corrects both of them. This conflict graph has at most $K \binom{d}{2}$ edges. It has an independent set of size at least

$$R \geq \frac{m^2}{m + Kd(d-1)}. \tag{1}$$

For completeness, order the vertices uniformly at random and retain those appearing before every neighbor. The retained set is independent, its expected size is $\sum_j (1 + \deg j)^{-1}$, and Cauchy–Schwarz gives (1).

On the retained columns each row corrects at most one column, and all mixed-moment identities persist. The one-column theorem therefore gives

$K \geq c_0 R^2$ for an absolute $c_0 > 0$. This estimate can be made valid also for $R = 1, 2$ by reducing c_0 , since $K \geq 1$. If $Kd(d-1) \leq m$, (1) gives $K \geq c_0 m^2/4$. Otherwise (1) and the same estimate give

$$K^3[d(d-1)]^2 \geq c_0 m^4/4.$$

The claimed bound follows. When $d = 1$ use the one-column theorem directly. $\square$

The argument uses neither monotonicity of the columns nor rationality. Distinctness and the interval constraint on the scalar baseline are needed for the invoked one-column theorem. Overlap growing with m is not excluded; the estimate is only a limitation on this sparse correction architecture.